

Weekly Sales Prediction Using Simple Linear Regression

Method

Five simple linear regression models were trained to predict Weekly_Sales, each using a single X variable. The data was split into 80% training and 20% testing (random_state=42), and the MSE was calculated on the test set.

A new feature, **Month**, was created from the Date column.

MSE Results

X Variable	MSE
Store	284,296,629,512
CPI	320,370,061,621
Temperature	320,515,110,459
Month	320,917,790,541
Fuel_Price	322,204,752,659

Conclusion

The best X variable is **Store**, as it has the lowest MSE (284,296,629,512).